

Personal Expense Tracker Bot

Checkpoint 2 — Phase 2 Complete

Overall Progress

Phase	Description	Status	Progress
Phase 1	Setup & Security	Complete	100%
Phase 2	Cloud Deployment	Complete	100%
Phase 3	Natural Language Logging	Planned	0%
Phase 4	Analytics Commands	Planned	0%
Phase 5	Dashboard & Visualization	Planned	0%
Phase 6	AI Features	Planned	0%

Phase 2 — Cloud Deployment (COMPLETE)

Objective: Move the bot from local laptop to cloud so it runs 24/7.

What Was Accomplished

- ✓ Migrated from Railway to Render (free tier)
- ✓ Fixed Google credentials corruption issue — regenerated service account key
- ✓ Removed credentials.json from entire git history using git-filter-repo
- ✓ Fixed .gitignore to exclude credentials.json and output.txt
- ✓ Added keep-alive HTTP server on port 8080 so Render does not kill the bot
- ✓ Resolved Telegram Conflict error by shutting down old Railway deployment
- ✓ Bot deployed successfully and running 24/7 on Render
- ✓ Tested /expense command — data saves correctly to Google Sheets
- ✓ Confirmed bot works with laptop completely off

Key Technical Details

Platform	Render.com (Free Web Service)
Region	Oregon, USA

Runtime	Python 3.12
Build Command	pip install -r requirements.txt
Start Command	python bot.py
Port	8080 (keep-alive server)
Service URL	https://personal-expense-tracker-e75z.onrender.com
GitHub Repo	github.com/wingFire-29/personal-expense-tracker
Google Sheet ID	1szlGWdA8GWq9jkzAm3ZGxFbbuaGNbXeIlFjQoA26fyM

Problems Solved in Phase 2

Problem: Invalid JWT Signature on Railway

Solution: Deleted old corrupted service account key and regenerated a fresh one from Google Cloud Console.

Problem: GOOGLE_CREDENTIALS returning None

Solution: Properly formatted credentials as single-line JSON and set as environment variable on Render.

Problem: Render killing bot (no open port)

Solution: Added a lightweight HTTP keep-alive server running on port 8080 using Python's built-in http.server.

Problem: GitHub blocking push (secret in history)

Solution: Used git-filter-repo to permanently remove credentials.json and output.txt from entire commit history.

Problem: Telegram Conflict error

Solution: Identified and deleted the old Railway deployment that was still running alongside Render.

Test Results

/start command	Passed
/expense 100 Food Lunch	Passed
Data saved to Google Sheets	Passed
Wrong format error handling	Passed
Bot works with laptop OFF	Passed

Upcoming Phases

Phase 3 — Natural Language Logging

- Type "spent 120 on lunch" instead of /expense 120 Food Lunch

- Claude AI auto-detects amount, category, and description
- Old /expense command still supported alongside natural language
- Uses Anthropic API for intelligent message parsing

Phase 4 — Analytics Commands

- /summary — overall expense summary
- /monthly — monthly breakdown by category
- /today — today's spending
- /budget — set and track budget limits

Phase 5 — Dashboard & Visualization

- Pie charts by category
- Monthly spending trends
- Daily average spending
- Visual insights from Google Sheets data

Phase 6 — AI Features

- Smart input: type "250 at Dominos" → auto-categorized as Food
- Automatic spending insights and alerts
- Budget recommendations based on spending patterns
- Full natural language processing